INFORM TANZANIA

Subnational Disaster Risk

From a hidden-sheet Excel template to a scalable, local-data risk engine

Prime Minister's Office – Disaster Management Department (PMO-DMD)

Reproducing the regional INFORM model exactly — and scaling it to all 195 councils

June 2026

BACKGROUND

What is INFORM?

- INFORM is a global, open composite index of disaster risk — a single 0–10 score per area (0 = lowest risk, 10 = highest).
- Tanzania runs the regional INFORM Model Template (the SADC / IGAD subnational adaptation). The official national INFORM risk is 4.1.
- **Risk is built from three dimensions — the INFORM logic:**
  - Hazard & Exposure — the threats (drought, flood, earthquake, conflict...) and who/what is exposed
  - Vulnerability — susceptibility of people (poverty, health, food insecurity, displacement)
  - Lack of Coping Capacity — weakness of institutions & infrastructure to respond (WASH, health, DRR, governance)
- **Each dimension is built from indicators → categories → dimension → the final risk index.**

THE MODEL TEMPLATE

Structure: 78 indicators → 6 categories → 3 dimensions → 1 risk

HAZARD & EXPOSURE	VULNERABILITY	LACK OF COPING CAPACITY
Natural (drought, flood, earthquake, landslide, cyclone, coastal, wildfire…) Human (conflict, violence)	Socio-economic (poverty, GDP, aid dependency, habitat) Vulnerable groups (health, nutrition, food security, displaced people)	Institutional (DRR, governance) Infrastructure (WASH, health access, communication, education)

78 raw indicators (53 currently used for Tanzania) → aggregated to 6 categories → 3 dimensions → one INFORM risk score, for every administrative unit.

HOW THE TEMPLATE WORKS

From an actual value to a 0–10 score — the standardisation chain

1. Actual value (natural unit)
2. Denominator (÷ population / area / GDP, or none)
3. Outlier cap — Tukey IQR fence MAX(MIN(x, Q3+1.5·IQR), Q1−1.5·IQR)
4. Transform — None | Logarithm LN(0.001+x) | Exponential
5. Min–max scaling — 10 × (x − Min) / (Max − Min) vs a fixed reference
6. Direction — if protective: 10 − score (higher = better)
7. **Clamp [0,10] → ROUND to 1 dp → the 0–10 indicator**

```
Excel:  =IF(val="","",IFERROR(MAX(0,MIN(10,ROUND(IF(sign="Decrease",10-base,base),1))),"No data"))
```

CAPTURING DIVERSITY

A few indicators — many different units (why standardisation is needed)

Indicator	Unit	Dimension
Food security – insufficient food	% of population (Phase 2+)	Vulnerability
Historic drought frequency	years	Hazard & Exposure
Internally displaced people	count (number of people)	Vulnerability
Gross National Income per capita	US$ (thousands, log)	Coping Capacity
Health expenditure per capita	US$ per capita	Coping Capacity
Physicians density	per 10,000 population	Coping Capacity
Measles incidence	per 1,000,000 population	Vulnerability
Soil erosion	Mg / ha / yr	Hazard & Exposure
Coastal erosion	metres / year	Hazard & Exposure

%, years, counts, US$, per-1,000, per-10,000, ratios, Mg/ha/yr, m/yr — all reconciled onto one 0–10 scale.

From standardised indicators to the INFORM risk

- **Component = AVERAGE of its indicators where Use = "Yes" (Excel AVERAGEIFS)**
- **Category = AVERAGE of its components (arithmetic mean)**
- **Dimension = scaled GEOMEAN of its categories**
  - ROUND((10 − GEOMEAN((10−c)/10×9+1)) /9 ×10 , 1)
- **INFORM Risk = ∛ (H × V × LCC) — cube-root of the three dimensions**
  - ROUND(H^(1/3) × V^(1/3) × LCC^(1/3) , 1)
- The geometric mean means a country cannot "average away" one very weak dimension — high hazard with low coping still yields high risk.

THE HONEST PROBLEM

The bottleneck: it all lives in deeply hidden Excel sheets

Every reference range, outlier fence, transform and aggregation is buried in hidden worksheets.

- Opaque — the math is invisible to the people who use and must trust the numbers.
- Fixed unit set — built for ~170 districts; it cannot scale to 195 councils, wards or villages.
- Manual & single-file — one workbook, no version control, easy to break, hard to audit.
- Closed to local data — no clean way to add a local indicator or feed council-level values.
- No automation — no live update, no link to early-warning or anticipatory action.
- **Result: the model stops at the national / regional level. Local risk — where decisions are made — is out of reach.**

THE ENABLER

We rebuilt the hidden sheets as a transparent engine — proven identical

- The hidden-sheet formulas were reverse-engineered and reproduced exactly in code (a standardisation + aggregation engine).
- **Proven byte-for-byte against the official workbook:**
  - 8,664 / 8,664 standardised indicator cells reproduce the template exactly
  - 170 / 170 district risk scores reproduce the template exactly
- Now it scales — the same engine runs unchanged on 195 councils and 31 regions (194/195 match district INFORM; the rest are new post-2024 councils).
- Spec-driven & open — add, delete or amend an indicator without touching the math; fully testable and version-controlled.
- **The regional INFORM core is never altered — it is reproduced exactly; everything new is additive (v2).**

CAPTURING REAL LOCAL DATA

Methods — a federated, bottom-up data system

- Federated collection tools — one shared Excel sent to every sector; they enter ACTUAL values in natural units and a locked formula shows the 0–10 live.
- **One accountable Tanzanian stakeholder per indicator:**
  - TMA (climate), GST (geology), NEMC (environment), MoH (health), NBS (statistics), MoA/MUCHALI (food), MoW (water), PMO-DMD (DRR)...
- Bottom-up aggregation — village → district → region → country; finer, well-filled data refines the higher level.
- Authentic sources used: NBS 2022 Census, TMA CHIRPS v3 / ERA5, GST seismic catalogue, MoH TDHS-MIS 2022, IPC/MUCHALI, TASAF, and satellite Earth Observation.
- **Earth Observation computed live (Earth Engine): MODIS NDVI, SMAP soil moisture, CHIRPS aridity — each validated against known geography before use.**

HONEST STATUS

Where this actually stands — no overselling

SOLID / EXACT

- Engine reproduces the template byte-for-byte: 8,664/8,664 cells, 170/170 risks
- Same engine runs at 195 councils / 31 regions (194/195 = district INFORM)
- 2022 census population per council (sum = 61,741,120, the exact national total)

PROTOTYPE / PROPOSED

- Advanced "exploded" multi-source indicators + weights = literature-proposed, NOT enforced
- EO drought layers (aridity P/PET, NDVI, soil moisture) computed & validated — research-grade
- Seasonality from CHIRPS 8-zone analysis — context layer, not yet in the index

TRIED — DID NOT WORK

- Sentinel-1 flood (VV<-17dB): FAILED validation — flagged soda-lakes/salt-flats as water → PULLED
- Lightning (LIS/OTD): not available on Earth Engine — not done
- SMAP soil moisture product is deprecated (ends 2022) — not current

STILL OPEN

- Many vulnerability indicators exist only at region/national level → council = labelled proxy
- No sustainable update pipeline; no live link to EOCC / early warning
- TASAF council poverty & IPC food: data exists but locked in PDFs/registry

THE GAP

What is still missing — the challenge ahead

- **Subnational risk is not yet operational — decisions happen at council/ward level, but official INFORM stops at national/regional.**
- Local data is scattered and uneven — many key indicators exist only at region or national level (poverty, health surveys, GDP).
- No sustainable pipeline — capturing, validating and updating local data is still manual and ad-hoc.
- Earth-Observation indicators (vegetation, soil moisture, flood, landslide) need proper computation, downscaling and validation.
- No live link to action — risk is not yet wired to the EOCC, early-warning, or LGA planning & budgeting.
- Capacity & financing gaps at the local-government level — tools, training and incentives to keep the data flowing.

OPPORTUNITIES

Research & financial support — for member-state adoption

RESEARCH

- EO computation & statistical downscaling to council/ward
- Validation against ground events & independent products
- Multi-source "exploded" indicators (one hazard, many institutions)
- Climate seasonality (8 rainfall zones) into risk & early warning
- Methods publishable as regional public goods

FINANCING & ADOPTION

- Fund local-data integration & the federated tools
- LGA capacity building — train sectors to capture & submit
- Scale to SADC / IGAD member states (transparent, open engine)
- Wire risk to PMO-DMD EOCC, anticipatory action & PO-RALG planning
- Tanzania as the subnational-INFORM pilot for the region

Summary

- The template math is fully understood and reproduced exactly (8,664/8,664; 170/170).
- It runs at council level (195) because it is code, not a hidden-sheet workbook.
- Local-data capture and EO are working but research-grade — and some attempts failed (stated plainly).
- Ask: research partnership + financing to integrate local data and support member-state adoption.